**10** Fig. 10.1 is a simplified diagram of a digital circuit. The output of logic gate Y controls a buzzer.

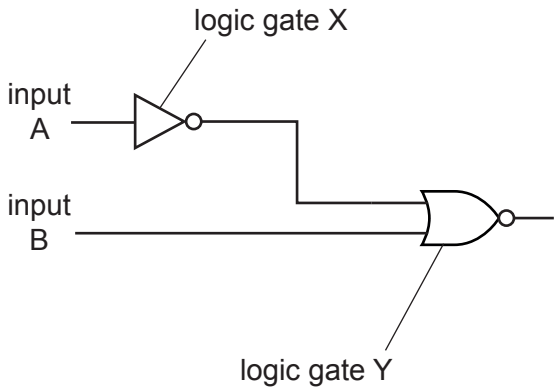



**Fig. 10.1**

**(a)** Complete Table 10.1, the truth table for the circuit.

**Table 10.1**

| input A | input B | output of X | output of Y |
|---|---|---|---|
| 0 | 0 | | |
| 0 | 1 | | |
| 1 | 0 | | |
| 1 | 1 | | |

[3]

**(b)** Input A is the output of a humidity sensor which gives logic 1 when the humidity is high and logic 0 when the humidity is low.

Input B is the output of a light sensor which gives logic 1 in bright light and logic 0 in darkness. The buzzer sounds when the output of Y is logic 1.

State the conditions of humidity and light when the buzzer is on.

..... [1]

**(c)** The output of the digital circuit alone is **not** able to operate the buzzer.

Ring the component from the list that must be connected between the output of the digital circuit and the buzzer.

**fuse** **heater** **relay** **resistor** **thermistor**

Explain your answer.

.....

.....

[3]

[Total: 7]